Unit: 6 Unit 4: Absolute Value **Name:** ____________

Topic: Absolute Value Inequalities and Their Applications **Date:** __________

Foundation (Jalapeno)

Q1. Solve the inequality $|x| < 3$.

(A) $-3 < x < 3$

(B) $-3leqxleq3$

(C) $0 < x < 3$

(D) $0leqxleq3$

(E) $x > 3$

Q2. What is the solution set in interval notation for $|x-2|geq1$?

(A) $(-infty, 1]cup[3, infty)$

(B) $(-infty, 1)cup(3, infty)$

(C) $[1, 3]$

(D) $(1, 3)$

(E) $(-infty, 1)cup[3, infty)$

Q3. Graph the solution set for $|x| > 2$ on a number line.

(A) $(-infty, -2)cup(2, infty)$

(B) $(-infty, -2]cup[2, infty)$

(C) $[-2, 2]$

(D) $(-2, 2)$

(E) $[-2, 2)$

Q4. A company packages coffee in bags that weigh within 0.5 pounds of the ideal weight of 2 pounds. What inequality represents the acceptable weights?

(A) $|x-2|leq0.5$

(B) $|x-2|geq0.5$

(C) $|x|leq0.5$

(D) $|x|geq0.5$

(E) $xleq2.5$

Q5. Solve the inequality $|3x-2| > 5$.

(A) $x < -1$ or $x > 2$

(B) $xleq-1$ or $xgeq2$

(C) $x > -1$ and $x < 2$

(D) $xgeq-1$ and $xleq2$

(E) $x = -1$ or $x = 2$

Q6. What is the solution set in interval notation for $|x+4|leq2$?

(A) $[-6, -2]$

(B) $(-6, -2)$

(C) $[-2, 6]$

(D) $(-2, 6)$

(E) $[-6, 2]$

Q7. Solve the inequality $|x|geq4$.

(A) $xleq-4$ or $xgeq4$

(B) $x < -4$ or $x > 4$

(C) $x = -4$ or $x = 4$

(D) $x > -4$ and $x < 4$

(E) $xgeq-4$ and $xleq4$

Q8. Graph the solution set for $|x-1| < 2$ on a number line.

(A) $(-1, 3)$

(B) $(-1, 3]$

(C) $[-1, 3]$

(D) $(-1, 3)$

(E) $[-1, 3)$

Open Ended Questions (FRQ)

Q9. Solve the inequality $|2x-5|geq3$ and express your answer in interval notation.

Q10. A temperature control system can maintain temperature within 2 degrees of the set point of 70 degrees. What inequality represents the acceptable temperatures? Solve the inequality and graph the solution set on a number line.

Chef's Tips

- Read the questions carefully.
- Use a number line to visualize the solution sets.
- Check your work and make sure to simplify your answers.